llm

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agents vs agentic ai -
While **AI agents** are individual, modular software entities designed to perform narrow, specific tasks using tools, **agentic AI** is the broader system architecture and operational paradigm where multiple agents, data sources, and reasoning loops are coordinated to manage complex, end-to-end workflows

Claude Code is the **agent harness**; when combined with a frontier Claude model, it creates an **agentic AI** software engineering system.

Agent Harness (The Execution & Control System): It is the runtime environment. It contains set of tools and the environmemt in which it executes. This includes the execution loop (perceive -> decide-> act ->observe), tool integrations (file editing, terminal commands, APIs), memory persistence, and security/permission checks.

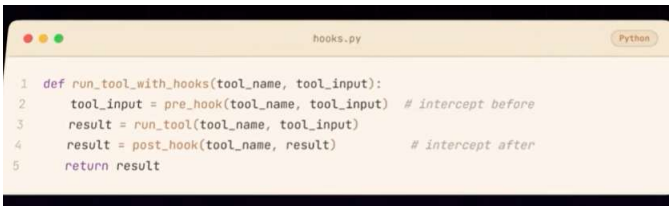
Agentic coding is the application of agentic AI principles specifically to software development—moving from passive code generation to autonomous, goal-driven software engineering

Agent harness + LLM = Agentic AI

Loop Phase	Primary System	Function
Observe	**Harness**	Captures raw environmental responses, errors, and status codes.
Perceive	**Harness -> LLM**	Harness turns observe raw data into context- suitable for the context window; Reads, parses, and comprehends the formatted context provided by the harness to update its internal context model.
Decide	**LLM**	Synthesizes context to select the next tool or text response.
Act	**Harness**	Intercepts tool calls and executes them in the external runtime.

1. **You** send a message ("Create a new file with a hello world function")
2. **The LLM** decides it needs a tool and responds with a structured tool call (or multiple tool calls)
3. **Your program** executes that tool call locally (actually creates the file)
4. **The result** gets sent back to the LLM
5. **The LLM** uses that context to continue or respond

```
# n0 master loop
while(tool_call):
  execute_tool()
  feed_results_to_model()
  repeat()
```



The harness spans subagent using another tool call. So spanning subagent is also just a tool call

The Anthropic Claude API separates inputs into **system** and **user** prompts to distinguish high-level developer instructions from runtime end-user input.

System Prompt: Defines the baseline rules, persona, formatting constraints, available tools, and guardrails set by the developer/operator. -- Also Prompt cacheing becomes better.
User propmt : What user entered.